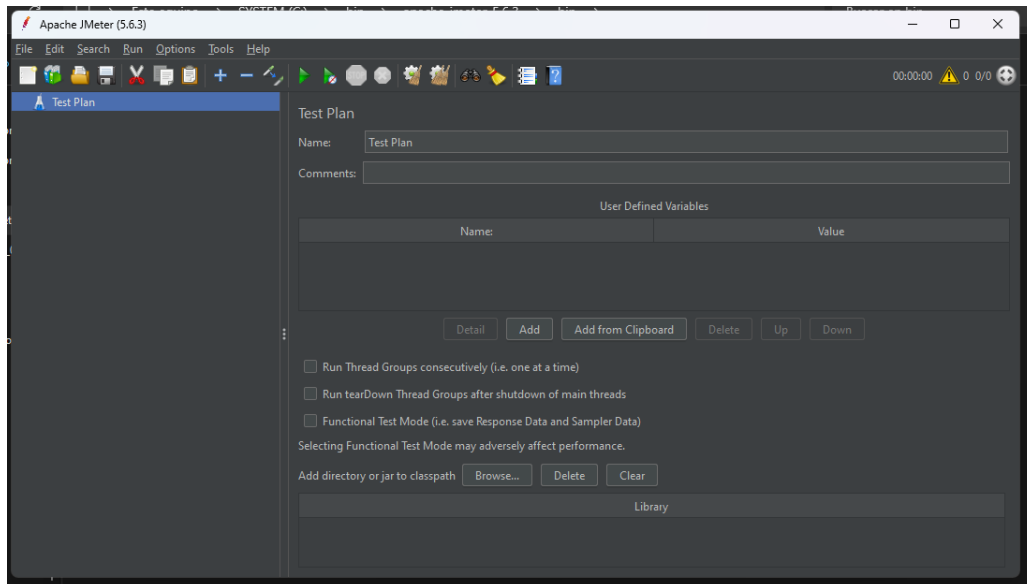


Nombre: RONALDINHO RODRIGUEZ ROMERO

Parte 1 — Instalación y verificación

1. Descargue Apache JMeter 5.6.x desde jmeter.apache.org (requiere Java 8 o superior).
2. Descomprima y ejecute bin/jmeter (Windows: jmeter.bat; Linux/Mac: ./jmeter).
3. Confirme que la API está levantada y respondiendo en localhost:3000.



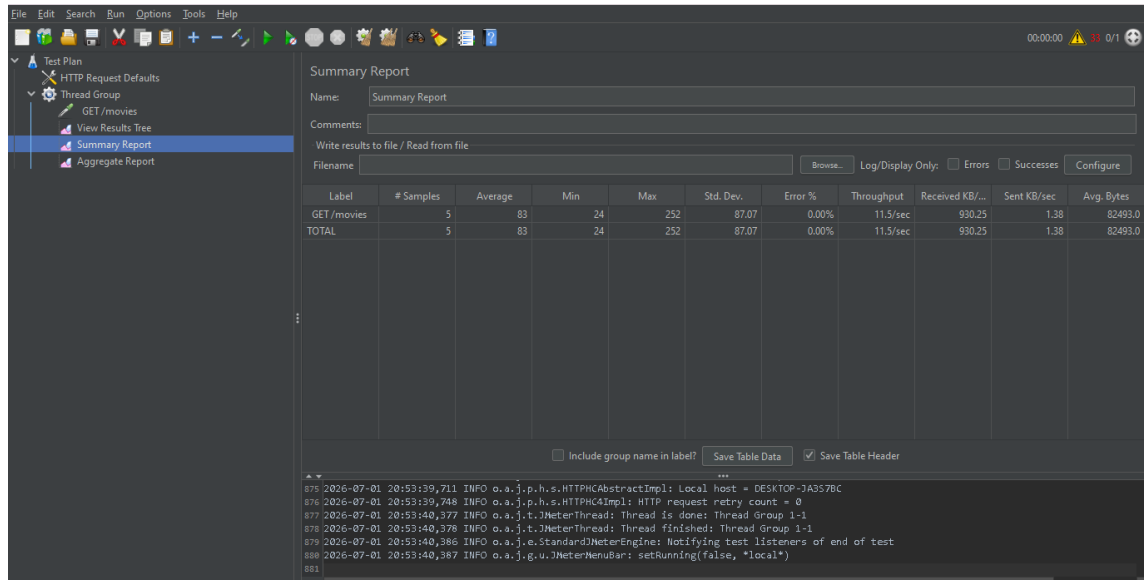
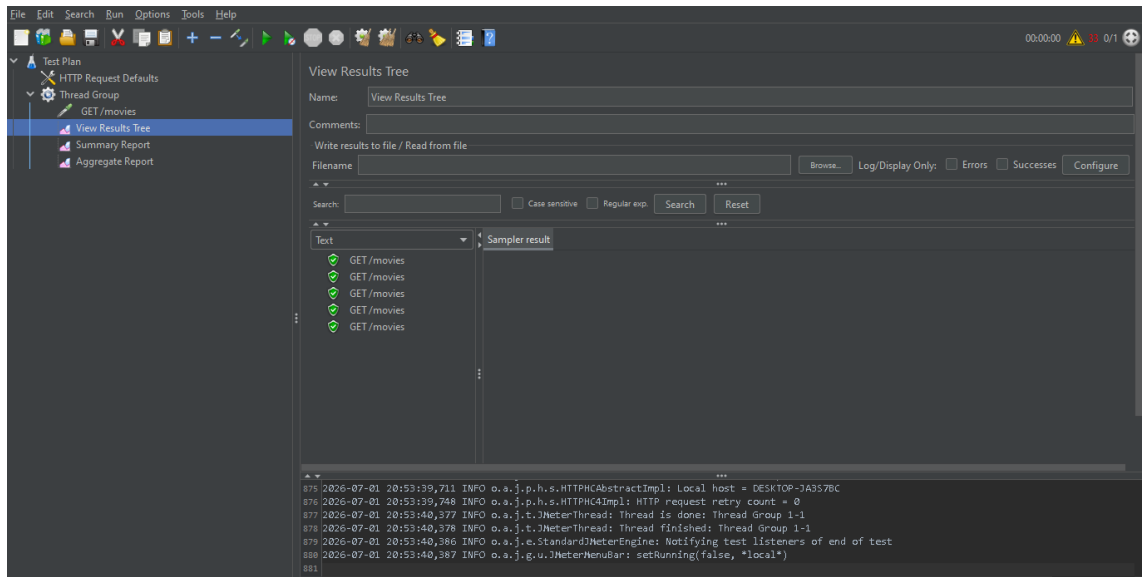
```
C:\Windows\System32\cmd.exe
Microsoft Windows [Versión 10.0.22631.3447]
(c) Microsoft Corporation. Todos los derechos reservados.

C:\bin\apache-jmeter-5.6.3\bin>java --version
java 26.0.1 2026-04-21
Java(TM) SE Runtime Environment (build 26.0.1+8-34)
Java HotSpot(TM) 64-Bit Server VM (build 26.0.1+8-34, mixed mode, sharing)

C:\bin\apache-jmeter-5.6.3\bin>
```

Parte 2 — Plan base (smoke test)

1. Cree un Test Plan y agregue un HTTP Request Defaults (Server Name: localhost, Port: 3000).
2. Agregue un Thread Group: 1 hilo, ramp-up 1 s, loops 5.
3. Agregue un HTTP Request GET /movies.
4. Agregue los listeners: View Results Tree, Summary Report y Aggregate Report.
5. Ejecute y verifique que todas las peticiones son exitosas (Error % = 0). Guarde como smoke.jmx.



The screenshot shows the JMeter Aggregate Report window. The left sidebar lists the test plan components: Test Plan, HTTP Request Defaults, Thread Group, GET /movies, View Results Tree, Summary Report, and Aggregate Report. The main panel displays the Aggregate Report for the 'GET /movies' request. The report includes a table with the following data:

Label	# Samples	Average	Median	90% Line	95% Line	99% Line	Min	Maximum	Error %	Throughput	Received K...	Sent KB/sec
GET /mov...	5	83	33	85	252	252	24	252	0.00%	11.5/sec	930.25	1.38
TOTAL	5	83	33	85	252	252	24	252	0.00%	11.5/sec	930.25	1.38

Below the table, there are checkboxes for 'Include group name in label?' and 'Save Table Data', and a checked checkbox for 'Save Table Header'. The bottom of the window shows a log of test execution details.

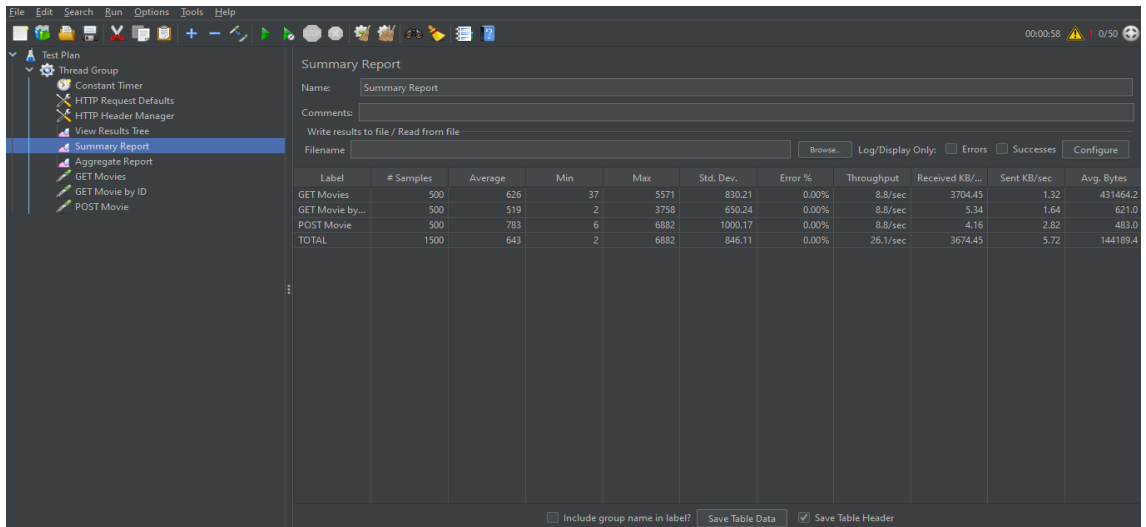
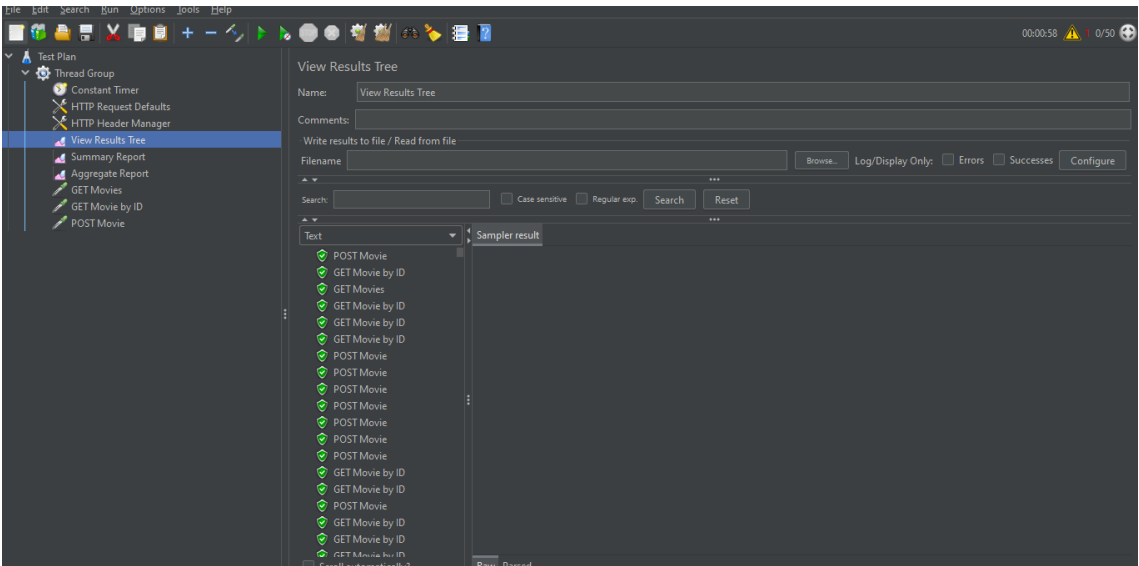
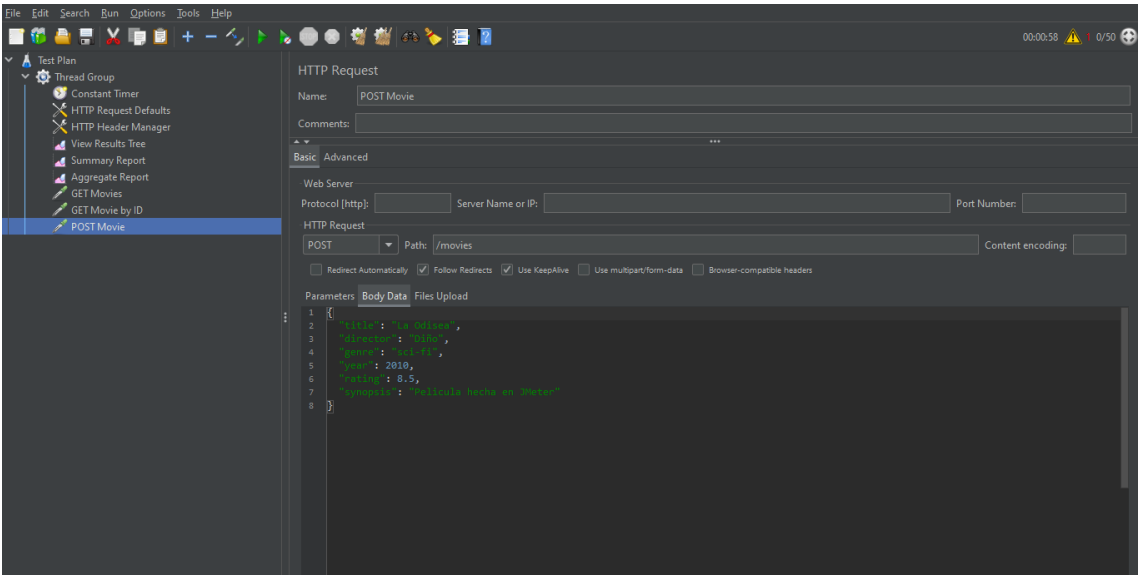
Parte 3 — Prueba de carga

1. Duplique el plan. Configure el Thread Group a 50 usuarios, ramp-up 30 s, loops 10.
2. Agregue al menos 3 peticiones distintas: GET /movies, GET /movies/{id} (con CSV Data Set Config de IDs) y POST /movies.
3. Agregue un Constant Timer o Gaussian Random Timer (p. ej. 300 ms) para simular usuarios reales.
4. Ejecute la prueba y registre throughput, media, percentil 90/95 y Error % de cada petición.
5. Guarde como carga.jmx y exporte los resultados a carga.jtl.

The screenshot shows the JMeter Thread Group configuration window. The left sidebar lists the test plan components: Test Plan, Thread Group, Constant Timer, HTTP Request Defaults, HTTP Header Manager, GET Movies, GET Movie by ID, POST Movie, View Results Tree, Summary Report, and Aggregate Report. The main panel displays the configuration for the 'Thread Group'.

Thread Group Configuration:

- Name: Thread Group
- Comments:
- Action to be taken after a Sampler error:
 - ☒ Continue
 - ☐ Start Next Thread Loop
 - ☐ Stop Thread
 - ☐ Stop Test
 - ☐ Stop Test Now
- Thread Properties:
 - Number of Threads (users): 50
 - Ramp-up period (seconds): 30
 - Loop Count: ☐ Infinite 10
 - ☒ Same user on each iteration
 - ☐ Delay Thread creation until needed
 - ☐ Specify Thread lifetime
 - Duration (seconds):
 - Startup delay (seconds):



Aggregate Report

Name: Aggregate Report

Comments:

Write results to file / Read from file

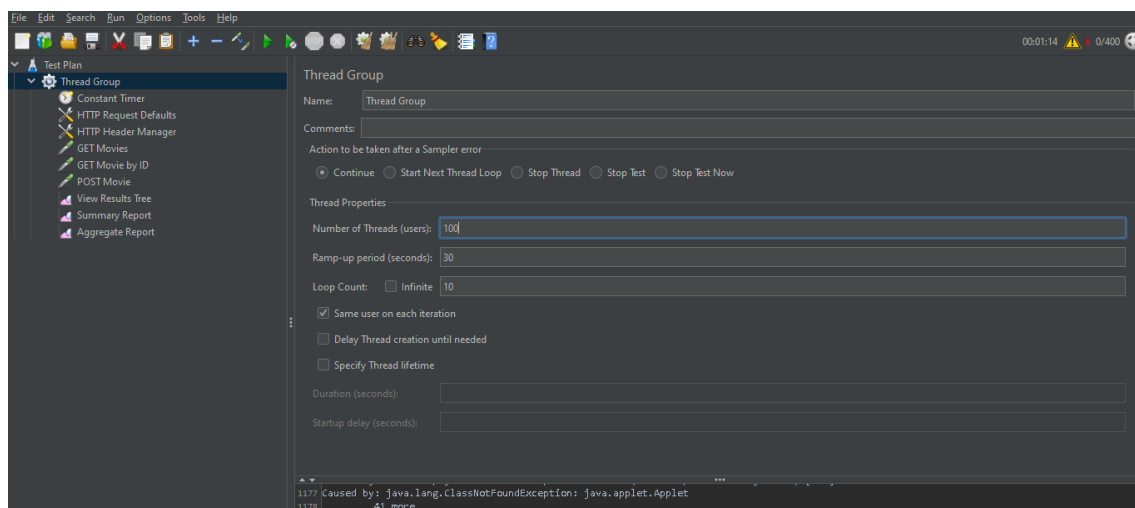
Filename: Browse... Log/Display Only: ☐ Errors ☐ Successes

Label	# Samples	Average	Median	90% Line	95% Line	99% Line	Min	Maximum	Error %	Throughput	Received K...	Sent KB/sec
GET Movies	500	626	393	1311	2135	4460	37	5571	0.00%	8.8/sec	3704.45	1.32
GET Movie...	500	519	284	1304	1887	2997	2	3758	0.00%	8.8/sec	5.34	1.64
POST Movie	500	783	473	1579	2718	5900	6	6882	0.00%	8.8/sec	4.16	2.82
TOTAL	1500	643	385	1415	2133	4289	2	6882	0.00%	26.1/sec	3674.45	5.72

Parte 4 — Prueba de estrés

1. Reutilice el plan de carga aumentando los usuarios en etapas: 100, 200 y 400 (puede usar varias corridas o el plugin Ultimate Thread Group).
2. Registre en qué nivel comienzan a aparecer errores o los tiempos de respuesta se disparan.
3. Identifique el punto de saturación (máxima carga con Error % aceptable, p. ej. < 1%).
4. Guarde el plan como `estres.jmx` y documente la tabla de resultados por nivel.

1



File Edit Search Run Options Tools Help

00:00:46 7 100/100

Test Plan

- Thread Group
 - Constant Timer
 - HTTP Request Defaults
 - HTTP Header Manager
 - GET Movies
 - GET Movie by ID
 - POST Movie
 - View Results Tree
 - Summary Report**
 - Aggregate Report

Summary Report

Name: Summary Report

Comments:

Write results to file / Read from file

Filename: Browse... Log/Display Only: ☐ Errors ☐ Successes

Label	# Samples	Average	Min	Max	Std. Dev.	Error %	Throughput	Received KB/...	Sent KB/sec	Avg. Bytes
GET Movies	227	4274	323	8913	2272.00	0.00%	5.0/sec	7230.88	0.75	1487800.0
GET Movie by...	200	4047	5	8536	2288.41	0.00%	4.4/sec	2.77	0.83	637.0
POST Movie	160	5197	267	11577	2311.41	0.00%	3.6/sec	1.70	1.15	482.0
TOTAL	587	4448	5	11577	2335.83	0.00%	12.9/sec	7234.15	2.69	575698.7

☐ Include group name in label? ☒ Save Table Header

```
1177 2026-07-01 22:23:25,872 INFO o.a.j.t.jMeterThread: Thread started: Thread Group 1-95
1178 2026-07-01 22:23:26,171 INFO o.a.j.t.jMeterThread: Thread started: Thread Group 1-96
1179 2026-07-01 22:23:26,475 INFO o.a.j.t.jMeterThread: Thread started: Thread Group 1-97
1180 2026-07-01 22:23:26,776 INFO o.a.j.t.jMeterThread: Thread started: Thread Group 1-98
1181 2026-07-01 22:23:27,076 INFO o.a.j.t.jMeterThread: Thread started: Thread Group 1-99
1182 2026-07-01 22:23:27,376 INFO o.a.j.t.jMeterThread: Thread started: Thread Group 1-100
1183
```

File Edit Search Run Options Tools Help

00:02:48 95/100

Test Plan

- Thread Group
 - Constant Timer
 - HTTP Request Defaults
 - HTTP Header Manager
 - GET Movies
 - GET Movie by ID
 - POST Movie
 - View Results Tree
 - Summary Report
 - Aggregate Report**

Aggregate Report

Name: Aggregate Report

Comments:

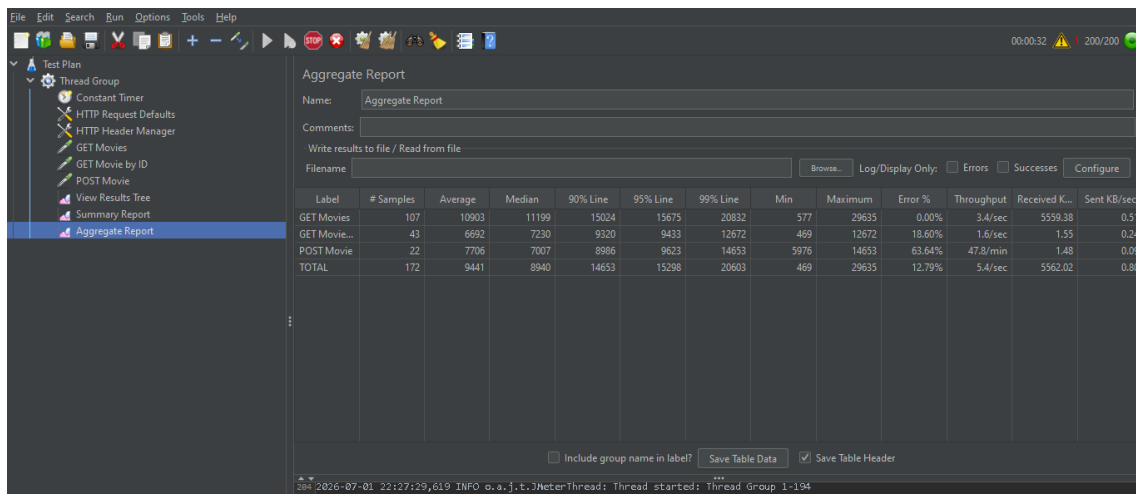
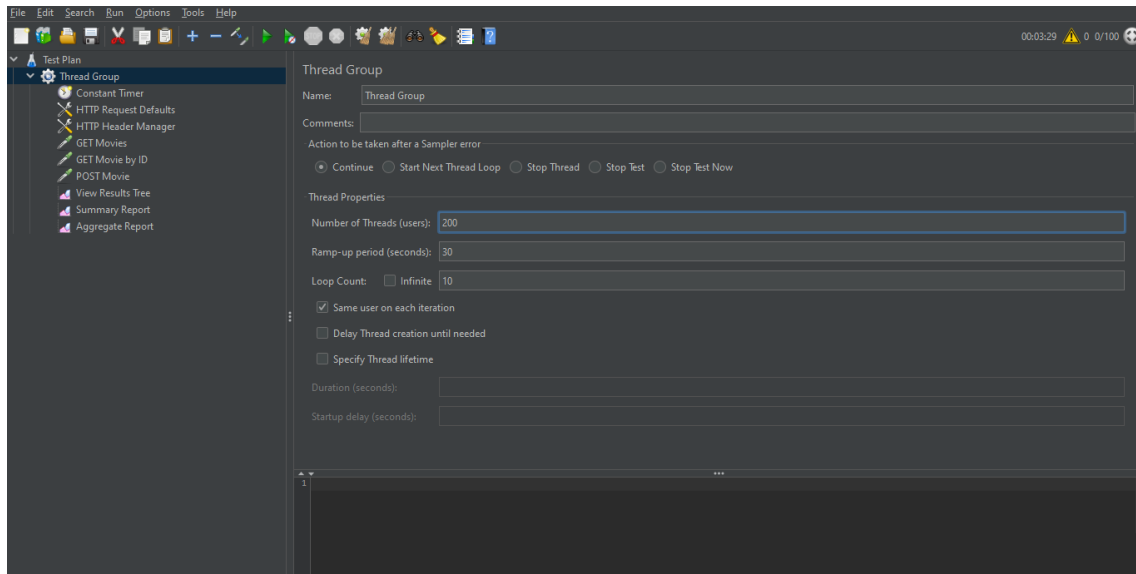
Write results to file / Read from file

Filename: Browse... Log/Display Only: ☐ Errors ☐ Successes

Label	# Samples	Average	Median	90% Line	95% Line	99% Line	Min	Maximum	Error %	Throughput	Received K...	Sent KB/sec
GET Movies	776	6398	6295	9835	10657	13862	323	14475	6.06%	4.6/sec	6585.37	0.66
GET Movie...	764	5911	5833	9535	10330	11354	5	14168	1.86%	4.6/sec	3.03	0.84
POST Movie	714	6884	6629	11150	11766	15287	267	17232	3.92%	4.3/sec	2.40	1.32
TOTAL	2254	6387	6283	10076	11105	13963	5	17232	3.99%	13.5/sec	6589.39	2.80

☐ Include group name in label? ☒ Save Table Header

```
1177 2026-07-01 22:25:43,649 INFO o.a.j.t.jMeterThread: Thread is done: Thread Group 1-2
1178 2026-07-01 22:25:43,650 INFO o.a.j.t.jMeterThread: Thread finished: Thread Group 1-2
1179 2026-07-01 22:25:44,593 INFO o.a.j.t.jMeterThread: Thread is done: Thread Group 1-6
1180 2026-07-01 22:25:44,593 INFO o.a.j.t.jMeterThread: Thread finished: Thread Group 1-6
1181 2026-07-01 22:25:44,597 INFO o.a.j.t.jMeterThread: Thread is done: Thread Group 1-8
1182 2026-07-01 22:25:44,597 INFO o.a.j.t.jMeterThread: Thread finished: Thread Group 1-8
```



3

